Placement-Aware Refinement for CLIP-Based Weakly Supervised Video Anomaly Detection

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***Abstract*—Fixed-length temporal aggregation in CLIP-based weakly supervised video anomaly detection preserves average appearance but suppresses local feature variation. Additional semantic supervision can also change classification predictions through their shared representation. We propose a placement-aware refinement framework that addresses these two issues by controlling where temporal evidence enters the model and where semantic optimization acts. First, pre-aggregation variation refinement extracts differences from raw CLIP snippet features and injects aligned variation statistics through a zero-initialized low-rank residual. A sparse-neighborhood refinement further strengthens similarity-based temporal propagation. Second, pathway-restricted semantic adaptation optimizes a semantic multiple-instance objective through the existing text-side adapter, preserving classification outputs by construction. On UCF-Crime and XD-Violence, the method improves the matched VadCLIP baseline by 0.28 AUC and 1.66 AP percentage points, respectively. Under the original VadCLIP training setting, it achieves 84.94% AP on XD-Violence in its best run. Component and placement ablations validate both refinements. The method adds 132,736 parameters and increases neural scoring latency by 3.3–3.6%, with no additional inference operations from semantic adaptation.**

***Index Terms*—weakly supervised video anomaly detection, vision-language models, temporal variation, semantic adaptation, intelligent surveillance**

I. Introduction

Intelligent surveillance systems must identify rare abnormal events in long video streams. Video-level labels indicate event presence without the expense of annotating temporal boundaries. Weakly supervised video anomaly detection (WSVAD) learns snippet-level anomaly scores from these labels through multiple-instance learning [1]. Frozen CLIP features [2] supply semantic information for lightweight temporal models. VadCLIP [3] combines classification and prompt-aligned semantic outputs within this setting.

Recent detectors improve feature discrimination, temporal context, and prompt learning [4]–[8]. An additional question concerns the *position* at which information and supervision enter the detector. Temporal modeling depends on the evidence retained in its input, while auxiliary objectives can change predictions that share updated parameters. Both dependencies matter when feature compression precedes a model with related outputs.

The first issue arises before temporal modeling. VadCLIP maps each training feature sequence to a fixed length. For long sequences, segment-mean aggregation reduces multiple consecutive snippets to one vector. Similar average appearances can result from very different local feature trajectories, so differencing segment means cannot recover the variation that occurred before aggregation. In the training features used here, 28.0% of UCF-Crime sequences and 30.4% of XD-Violence sequences undergo this averaging. Extracting the variation beforehand provides information that the temporal model would otherwise have to infer from compressed appearance features.

The second issue concerns the optimization of the semantic output. In VadCLIP, classification scores weight a pooled video representation that conditions the text prompts. Both outputs also use a shared temporal representation. Applying an additional semantic objective to the entire network can therefore alter the classification output. The text-side adapter, however, lies downstream of the classification computation. This dependency offers a direct way to improve prompt-aligned predictions while keeping an established classifier unchanged.

We address these issues with the placement-aware framework in Fig. 1. Stage 1 computes absolute differences of raw CLIP snippet features before aggregation and fuses the aligned variation statistics through a zero-initialized low-rank residual. It also refines similarity propagation by mixing the original adjacency with a sparse nearest-neighbor term. Stage 2 adds a semantic multiple-instance objective and updates only VadCLIP's existing text-side adapter. The common principle is to preserve temporal evidence before it is compressed and to place auxiliary optimization where it can adapt semantics without moving classification scores.

Component, placement, and objective ablations validate the framework's design. Our contributions are:

- **Pre-aggregation variation refinement.** We introduce an aligned temporal residual that supplies explicit snippet-variation statistics to the detector before temporal modeling. Component and pre/post-aggregation comparisons establish the value of this design, complemented by sparse-neighborhood graph refinement.
- **Classifier-preserving semantic adaptation.** We derive the update restriction from VadCLIP's computation graph and apply semantic multiple-instance supervision to an existing text-side adapter. Semantic predictions can adapt while classification outputs remain unchanged, without adding inference operations.
- **Effective, lightweight refinement.** Experiments on UCF-Crime and XD-Violence show improvements of 0.28 AUC and 1.66 AP points over matched VadCLIP retraining. The original training setting yields a best-run XD-Violence AP of 84.94%, and neural scoring latency increases by less than 4%.

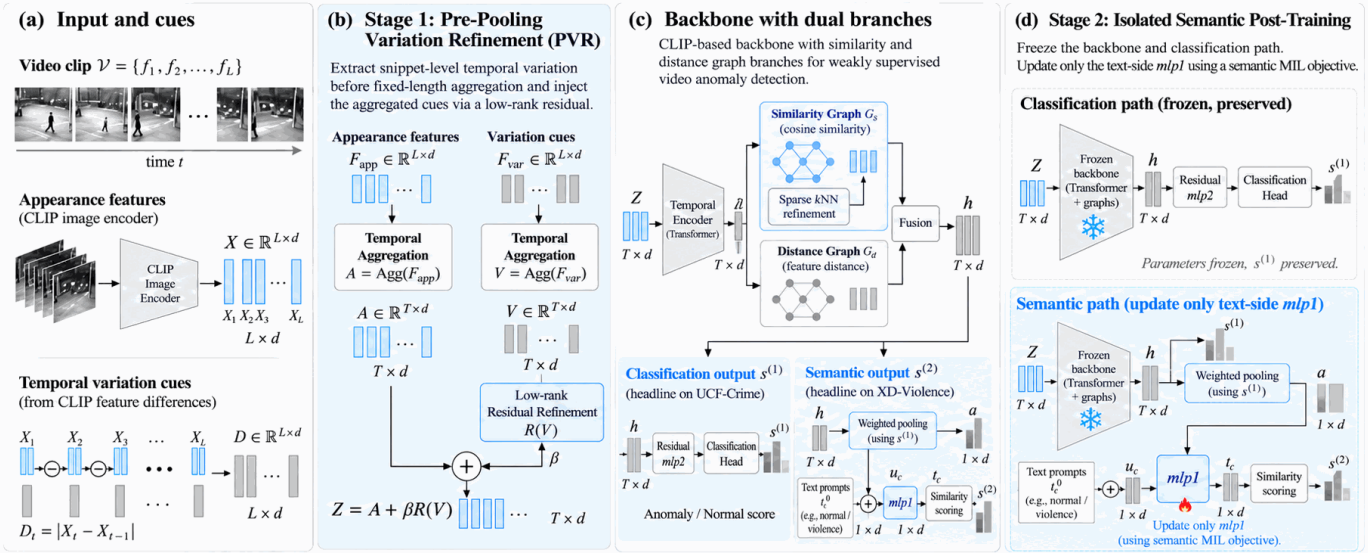


Fig. 1. Overview of placement-aware refinement. (a) Raw CLIP snippet features and temporal variation. (b) Pre-aggregation variation refinement with aligned aggregation and low-rank residual fusion. (c) Similarity and distance graph branches and two prediction pathways. (d) Restricted semantic post-training updates only the text-side adapter while preserving classification outputs.

II. Related Work

A. Weakly Supervised Video Anomaly Detection

The multiple-instance formulation of Sultani et al. [1] supports temporal anomaly scoring from video-level supervision. Subsequent work improves the evidence used for scoring: graph-based label cleaning addresses noisy snippet labels [9], RTFM learns discriminative feature magnitudes [4], MGFN combines magnitude contrast with context modeling [10], and UR-DMU regulates uncertainty through dual memories [5]. HL-Net models similarity and temporal-distance relations for the XD-Violence benchmark [11]. Our method works with frozen snippet features and targets the interface between variable-length sequences and fixed-length temporal learning.

B. Vision-Language Models for Anomaly Detection

CLIP-TSA applies temporal self-attention to CLIP features [12], while PEL4VAD integrates prompt-enhanced context learning [6]. VadCLIP adds prompt-aligned semantic prediction to the classification pathway [3]. Normality-guided prompts [7] and spatio-temporal prompts [8] further enrich anomaly semantics. VadCLIP++ learns dynamic text prompts using adjacent-frame difference information and combines them with spatial feature selection [13]. Its use of temporal differences concerns dynamic prompt learning. Our temporal refinement instead computes variation before feature aggregation, and our semantic refinement controls which existing parameters an auxiliary objective may update. Language-guided methods also support training-free anomaly scoring [14] and multimodal anomaly reasoning [15]. We focus on a lightweight, trainable extension of a snippet-scoring detector.

C. Temporal Cues and Lightweight Adaptation

Explicit temporal cues are well established in video understanding. TDN uses frame differences [16], TSM exchanges temporal information through channel shifts [17], and X-CLIP introduces cross-frame communication [18]. Here, the input is already encoded by a frozen backbone, making the ordering of feature differencing and temporal aggregation the relevant design choice. Graph convolution [19] provides the relation-propagation mechanism refined by our kNN mixture.

Adapters [20], low-rank updates [21], prompt learning [22], CLIP feature adapters [23], and ST-Adapter [24] demonstrate efficient adaptation of pretrained representations. Our two stages use this efficiency in different ways: a small residual introduces missing temporal statistics, whereas semantic adaptation reuses an existing adapter. The latter is selected by its downstream dependency, which makes classification preservation a structural property rather than a consequence of balancing loss weights.

III. Method

A. Dual-Output Detection Pipeline

Let $X \in \mathbb{R}^{L\times d}$ denote a video's raw CLIP snippet features, where L is variable and $d = 512$. During training, Agg maps each sequence to $T = 256$ positions: sequences longer than T are averaged over uniform temporal segments, and shorter sequences are zero-padded. VadCLIP's local temporal transformer, similarity graph branch, and distance graph branch transform the input into a shared representation $h \in \mathbb{R}^{T\times d}$ [3]. The classification logits are

$$s^{(1)} = f_{\mathrm{cls}}\big(h + \mathrm{mlp}_2(h)\big). \tag{1}$$

The semantic pathway uses these logits to form a normalized video vector and condition each class prompt:

$$a = \mathrm{norm}\left(\sum_j s_j^{(1)} h_j\right), \tag{2}$$

$$u_c = t_c^0 + a, \qquad t_c = u_c + \mathrm{mlp}_1(u_c), \tag{3}$$

$$s_{j,c}^{(2)} = \frac{\cos(h_j, t_c)}{\tau}, \qquad p_j = 1 - \mathrm{softmax}\left(s_j^{(2)}\right)_{\mathrm{normal}}, \tag{4}$$

where norm is ℓ_2 normalization, t_c^0 is the encoded class prompt, and $\tau = 0.07$. The classification output in Eq. (1) is independent of mlp_1, while the semantic output depends on classification through a. This directed dependency is central to Stage 2. Following VadCLIP, the headline metrics use $s^{(1)}$ on UCF-Crime and p on XD-Violence.

B. Pre-Aggregation Variation Refinement

Aligned temporal evidence. For a long sequence, let $\mathcal{B}_j$ be the raw snippet indices assigned to aggregated position j. The appearance input is the segment mean $A_j = |\mathcal{B}_j|^{-1} \sum_{t \in \mathcal{B}_j} X_t$. To retain explicit variation statistics, we first compute

$$D_1 = 0, \qquad D_t = |X_t - X_{t-1}|, \quad t = 2, \ldots, L, \tag{5}$$

with elementwise absolute value. We then aggregate both streams using the same boundaries:

$$A = \mathrm{Agg}(X), \qquad V = \mathrm{Agg}(D). \tag{6}$$

Thus V_j summarizes raw feature changes assigned to segment j, including a boundary transition when its first snippet follows another segment. Concatenating the appearance and variation channels before temporal processing keeps their positions aligned.

This ordering extracts information that differences of segment means do not specify. For example, a constant feature trajectory and an alternating trajectory around the same center can have identical means but different average absolute changes. After aggregation, a temporal difference sees the means alone. Computing Eq. (5) first exposes the variation statistic to the detector alongside appearance. In our long training sequences, aggregation averages 5.76 snippets per segment on UCF-Crime and 2.23 on XD-Violence. On shorter sequences, pre- and post-aggregation differences coincide at valid positions before padding.

Residual integration. The aligned variation stream is mapped back to the appearance space with a low-rank residual:

$$R(V) = W_2\, \mathrm{GELU}(W_1\, \mathrm{LN}(V)), \qquad Z = A + \beta R(V), \tag{7}$$

where the projection widths are $512 \to 128 \to 512$, biases are implicit, and $\beta = 0.2$. The second projection and its bias are zero-initialized, so the input residual initially vanishes: $Z = A$. This initialization lets the model learn a variation correction from the original appearance input. Z is injected before positional embeddings and the temporal transformer, allowing both prediction pathways to use the refined representation. The residual adds 132,736 parameters.

At inference, raw appearance and variation features are split into the same non-overlapping 256-snippet windows, with padding in the final window. No segment averaging is performed. Variation is computed over the full raw sequence before windowing, preserving transitions across adjacent windows in the variation input. Training and inference therefore share the feature-space difference and residual mapping while retaining VadCLIP's windowed evaluation.

C. Sparse-Neighborhood Graph Refinement

We strengthen similarity-based propagation with a parameter-free nearest-neighbor mixture. Let F be the transformer output and $S_{ij} = \cos(F_i, F_j)$. VadCLIP thresholds similarities at 0.7 before applying a row-wise softmax [3]. A thresholded zero still contributes e^0 to the softmax denominator, so these entries retain nonzero weights. A sparse neighborhood term concentrates part of the propagation mass on the most similar snippets:

$$G_{\mathrm{kNN}} = \mathrm{softmax}\big(M_k(S)\big), \qquad G = (1-\alpha)G_{\mathrm{orig}} + \alpha G_{\mathrm{kNN}}, \tag{8}$$

where M_k retains the k largest similarities per valid row, including self, and masks the others to $-\infty$. We use $k = 16$ and $\alpha = 0.25$. The mixture retains global context through G_{orig} and increases the contribution of the nearest-neighbor set through G_{kNN}. Both terms are row-normalized on valid positions. The final mixture retains a dense component; the sparse term changes the allocation of propagation weights without introducing trainable parameters. Only the similarity branch uses G. The distance branch and the concatenation/projection that produce h remain as in VadCLIP.

D. Pathway-Restricted Semantic Adaptation

Semantic multiple-instance objective. VadCLIP supervises prompt-aligned logits with its class-level CLASM objective [3]. We supplement it with a binary anomaly objective applied to the semantic probability p in Eq. (4). For valid sequence length ℓ, define $k_b = \lfloor \ell/16 \rfloor + 1$ and let $\mathcal{K}$ contain the k_b highest-scoring valid positions. With video-level anomaly label y, the objective is

$$q = \frac{1}{k_b} \sum_{j \in \mathcal{K}} p_j, \qquad \mathcal{L}_{\mathrm{S2}} = \mathcal{L}_{\mathrm{CLASM}} + \lambda\ \mathrm{BCE}(q, y), \tag{9}$$

where $\lambda = 0.1$. CLASM retains class-level alignment, while the binary term directly supervises the anomalousness represented by p. Both use the original video-level labels.

Classification-preserving update. Stage 2 starts from the Stage 1 checkpoint and optimizes Eq. (9) with respect to θ_{mlp_1} alone. All other parameters are frozen, including the temporal network, classification head, CLIP encoders, and prompt embeddings. The frozen components run in evaluation mode. Since mlp_1 is downstream of the classifier,

$$\frac{\partial s^{(1)}}{\partial \theta_{\mathrm{mlp}_1}} = 0. \tag{10}$$

For a fixed input, h, $s^{(1)}$, a, and t_c^0 are therefore unchanged during this stage. Semantic predictions can still adapt through

TABLE I
PUBLISHED CLIP-BASED BENCHMARK RESULTS (%). EARLIER COMPARISON ROWS FOLLOW VADCLIP [3]; VADCLIP++ FOLLOWS [13]. OUR OFFICIAL-SETTING RESULT IS THE BEST OF THREE RUNS.

Method	UCF AUC	XD AP
Sultani et al. [1]	84.14	75.18
HL-Net [11]	84.57	80.00
RTFM [4]	85.66	78.27
UR-DMU [5]	86.75	82.41
CLIP-TSA [12]	87.58	82.17
VadCLIP [3]	88.02	84.51
VadCLIP++ [13]	88.12	85.03
Ours (official setting)	–	**84.94**

$t_c = u_c + \mathrm{mlp}_1(u_c)$. This restriction separates semantic optimization from classification preservation without requiring a compensating classification loss or a choice of trade-off weight between the two outputs. It preserves the classifier established by Stage 1, including any downstream thresholding of its scores. Stage 2 updates 2,099,712 existing parameters and adds no inference operations or parameters.

IV. EXPERIMENTS

A. Experimental Setup

Datasets and metrics. We evaluate on UCF-Crime [1] and XD-Violence [11] using VadCLIP's released CLIP features, dataset splits, and frame-level evaluation [3]. Their test sets contain 290 and 800 videos, respectively. The headline metrics are frame-level AUC from the classification output on UCF-Crime and frame-level AP from the semantic output on XD-Violence. We also evaluate both outputs on UCF-Crime and classification AP on XD-Violence. Scores are percentages; improvements are percentage points.

Training. Stage 1 retains the original dataset-specific configuration and losses for 10 epochs. UCF-Crime uses two temporal transformer layers, attention window 8, learning rate 2×10^{-5}, and batch size 64. XD-Violence uses one layer, window 64, learning rate 10^{-5}, and batch size 96. Stage 2 uses three epochs of AdamW with learning rate 2×10^{-5}, weight decay 0.01, and the same dataset-specific batch sizes. The method hyperparameters $\beta = 0.2$, $k = 16$, $\alpha = 0.25$, and $\lambda = 0.1$ are shared across datasets.

Evaluation settings. For controlled comparisons and ablations, we evaluate the final checkpoint after the fixed training budget; test data are not used in training or checkpoint selection. Runs use seeds 234–236, with the same seed and sample order for each comparison. The component study with PVR alone uses the available matched seeds 234/235. Separately, the XD-Violence official-setting experiment retains the original VadCLIP script's epoch-wise test-AP checkpoint selection and reload procedure. Table I reports its best run, and Table III provides the matched runs.

B. Detection Performance

Published benchmark comparison. Table I compares representative CLIP-feature results. Our method reaches **84.94% AP** on XD-Violence under the original VadCLIP training setting, 0.43 points above the published VadCLIP result of 84.51%. The corresponding retrained baseline reaches 84.21%, giving a same-seed improvement of 0.73 points. Across the three official-setting runs in Table III, mean AP increases from 82.91% to 83.63%, with a paired gain of 0.73 ± 0.16 points. The method improves the baseline in each run. This comparison demonstrates that the refinement is effective within the training configuration of the method it extends.

Controlled comparisons. Table II evaluates the complete two-stage model against retrained VadCLIP with the fixed-epoch setting. UCF-Crime AUC increases from 86.25% to 86.53%, and XD-Violence AP from 77.57% to 79.23%. The respective paired improvements are 0.28 ± 0.15 and 1.66 ± 0.34 points, positive in all three runs. The table also exposes the effect beyond the headline UCF metric: classification AP improves by 0.78 points, semantic AUC by 0.44, and semantic AP by 0.85. Thus, the UCF improvement is reflected in the mean ranking metrics of both outputs.

On XD-Violence, classification AP is 79.54% for the complete method and 79.61% for the baseline, while semantic AP provides the principal improvement. This distinction matches the design of the framework: Stage 1 changes the representation read by both outputs, and Stage 2 targets semantic scores while preserving the Stage 1 classifier. The following ablations identify the contribution of each operation.

C. Component and Mechanism Ablations

PVR independently improves the detector. Table IV(a) compares VadCLIP, PVR alone, and PVR with the kNN mixture using the same two seeds. PVR raises UCF-Crime AUC from 86.44% to 86.69%, a gain of 0.25 points without graph refinement. Adding the kNN term further increases AUC to 86.80%. Its incremental gains over PVR are 0.16 and 0.06 points for the two seeds. This component study establishes the standalone contribution of temporal variation and the additional benefit of refining its similarity-based propagation.

Pre-aggregation placement matters. Table IV(b) replaces the raw-sequence variation with $D'_j = |A_j - A_{j-1}|$ while retaining the residual, graph mixture, and remaining settings. Pre-aggregation variation achieves 86.53% AUC compared with 85.86% for post-aggregation variation, a 0.67 ± 0.05 point advantage. Because the added module is otherwise the same, this comparison identifies pre-aggregation placement as a material design choice. The valid inputs coincide on short, padded training sequences; the variants differ on averaged training sequences and at test-window boundaries. The result is consistent with the intended role of supplying explicit raw-sequence variation to the temporal model.

The semantic objective improves adaptation. The equal-budget control C in Table IV(c) updates the same mlp_1 parameters for three epochs from the same Stage 1 checkpoint, but uses CLASM alone ($\lambda = 0$). Adding the binary semantic objective increases AP from 77.59% to 79.23%, a paired gain of 1.64 ± 0.43 points. This isolates the contribution of anomaly supervision from the additional optimization budget.

TABLE II

FIXED-EPOCH RESULTS (MEAN ± STANDARD DEVIATION, %) FOR THE SAME SEEDS 234–236. SUBSCRIPTS IDENTIFY THE CLASSIFICATION (1) AND SEMANTIC (2) OUTPUTS. Δ IS THE PAIRED IMPROVEMENT OVER RETRAINED VADCLIP.

	UCF-Crime				XD-Violence	
Method	AUC_1	AP_1	AUC_2	AP_2	AP_2	AP_1
VadCLIP	86.25 ± 0.83	29.26 ± 3.98	84.57 ± 0.89	26.46 ± 1.20	77.57 ± 2.19	79.61 ± 0.86
Ours	**86.53 ± 0.85**	**30.04 ± 2.81**	**85.01 ± 1.06**	**27.30 ± 1.60**	**79.23 ± 1.87**	79.54 ± 0.97
Paired Δ	+0.28 ± 0.15	+0.78 ± 1.16	+0.44 ± 0.32	+0.85 ± 0.42	+1.66 ± 0.34	−0.06 ± 0.11

TABLE III

XD-VIOLENCE AP (%) USING THE ORIGINAL VADCLIP TRAINING SETTING. EACH ROW PAIRS THE SAME SEED; Δ DENOTES OURS MINUS VADCLIP.

Seed	VadCLIP	Ours	Δ
234	84.21	**84.94**	+0.73
235	81.77	**82.65**	+0.88
236	82.74	**83.31**	+0.57
Mean	82.91 ± 1.23	83.63 ± 1.18	+0.73 ± 0.16

TABLE IV

COMPONENT AND MECHANISM ABLATIONS (%). PANEL (A) USES SEEDS 234/235; PANELS (B,C) USE 234–236. ALL VALUES ARE MEANS. Δ IS RELATIVE TO THE BASELINE WITHIN EACH PANEL. C AND FULL STAGE 2 SHARE THE CHECKPOINT, SAMPLE ORDER AND THREE-EPOCH BUDGET, DIFFERING ONLY IN λ.

(a) Stage 1 components: UCF AUC_1		
Variant	AUC	Δ
VadCLIP	86.44	–
+ PVR	86.69	+0.25
+ PVR + kNN	86.80	+0.36
(b) Variation placement: UCF AUC_1		
Variant	AUC	Δ
VadCLIP	86.25	–
Post-aggregation + kNN	85.86	-0.39
Pre-aggregation + kNN	86.53	+0.28

(c) Semantic adaptation: XD-Violence			
Variant	AP_2	Δ	AP_1
VadCLIP	77.57	–	79.61
Stage 1	77.44	-0.14	79.54
Stage 1 + CLASM (C)	77.59	+0.02	79.54
Stage 1 + Stage 2	79.23	+1.66	79.54
VadCLIP + Stage 2	79.46	+1.89	79.61

Applying Stage 2 directly to VadCLIP also improves semantic AP, reaching 79.46%. Together, the controls establish semantic adaptation as the source of the XD-Violence gain and PVR as an independently effective refinement on UCF-Crime.

The classifier is preserved during Stage 2. The classification AP in Table IV(c) remains 79.54% for the Stage 1, control C, and full Stage 2 checkpoints. Fixed-input checks during Stage 2 additionally record zero change in classification logits and no changed parameter outside mlp_1. These observations agree with Eq. (10): the added objective adapts the semantic pathway without modifying the classifier that supplies the conditioning weights.

TABLE V

NEURAL SCORING COST FOR ONE 256-SNIPPET WINDOW ON AN RTX 4060 LAPTOP GPU. PARAMETERS EXCLUDE FROZEN CLIP ENCODERS; PEAK ALLOCATED MEMORY IS IN MIB.

Data	Model	Params (M)	Time (ms)	Memory
UCF	VadCLIP	12.642	28.88	670.05
	Ours	12.775	29.84	671.05
XD	VadCLIP	9.490	17.44	643.25
	Ours	9.623	18.06	644.26

D. Efficiency and Practical Integration

Table V reports measurements on an NVIDIA GeForce RTX 4060 Laptop GPU. Each timing is the median of 200 forward passes after 20 warm-up passes, with CUDA synchronization. The measured network takes precomputed appearance and variation tensors and includes prompt encoding; visual feature extraction and variation preprocessing are outside the timed region.

Stage 1 adds 132,736 parameters. Neural scoring latency increases from 28.88 to 29.84 ms per UCF-Crime window and from 17.44 to 18.06 ms per XD-Violence window. Using the unrounded measurements, these increases are 3.3% and 3.6%. Peak allocated memory increases by approximately 1.01 MiB. Stage 2 only changes the values of an existing adapter, so it adds no inference parameters or operations to the Stage 1 model.

The framework separates temporal representation refinement from semantic adaptation in a way that supports integration into a video-analysis pipeline. PVR supplies aligned variation features without replacing the CLIP encoders or prediction heads. Semantic adaptation can then improve prompt-aligned scores while preserving classification-based alerts associated with the Stage 1 checkpoint. This property is useful when anomaly scores feed an established monitoring or human-review workflow: semantic refinement does not itself require recalibrating thresholds on the unchanged classification output. The experiments demonstrate this output-preserving property and its low scoring overhead within VadCLIP.

V. Conclusion

We presented placement-aware refinement for CLIP-based weakly supervised video anomaly detection. Pre-aggregation variation refinement exposes raw snippet dynamics to the temporal model through aligned aggregation and a low-rank residual, complemented by a nearest-neighbor graph mixture. Pathway-restricted semantic adaptation adds anomaly supervision to an existing text-side adapter while preserving classification scores. Component, placement, and equal-budget ablations establish the roles of these designs. The complete method improves matched VadCLIP results on UCF-Crime and XD-Violence, reaches 84.94% best-run AP under the original XD-Violence training setting, and adds less than 4% neural scoring latency. These results support a lightweight extension that improves temporal evidence and semantic prediction through targeted computational placement.

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